



Introduction

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This content applies to Cumulocity and to all subsequent releases.

Specifications contained herein are subject to change and these changes will be reported in subsequent versions.

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APPLICATION ENABLEMENT

Cumulocity provides a comprehensive application enablement platform that combines no-code tools with advanced development capabilities, enabling users to build and deploy IoT solutions that are tailored to business requirements.

The Cumulocity flexible architecture supports different levels of application enablement:

- **No-code enablement:** Allows users to quickly configure existing applications and extend functionalities through available extensions without the need to write code.
- **Low-code enablement:** Enables the creation of custom logic using intuitive tools like Analytics Builder and pre-built components, making it easy for users with minimal coding experience to develop solutions.
- **Code-based enablement:** For full customization, developers have access to comprehensive SDKs to build tailored solutions, offering maximum flexibility and control.

SELF-SERVICE APPLICATIONS

For users with no or minimal coding experience, Cumulocity includes:

- **Cockpit application** - Offers no-code tools that enable users to monitor and visualize IoT data quickly. It features a drag-and-drop interface, a pre-built [widget library](#) and [smart rules](#) for creating [dashboards](#) and analytics. This allows business users to implement real-time monitoring and data visualization solutions without extensive coding knowledge.
- **Digital Twin Manager** - Helps users create and manage blueprints of their assets, including their properties, relationships, and connections to sensor and device data.
- **Branding manager** - Enables organizations to modify logos, colors, fonts, and [domain names](#) for a consistent, white-labeled experience that aligns with their brand identity.
- **Extensions library** - Offers pre-built official & community-supported, open-source plugins that support adding additional functionalities in the platform that can be integrated seamlessly into applications.

CUSTOMIZATION & DEVELOPER TOOLS

For more advanced customization, Cumulocity provides comprehensive development capabilities:

- **Web SDK** - Provides all necessary clients and components to quickly build frontends in Cumulocity and fully utilize the Microfrontend architecture capabilities. Developers can start with a simple UI plugin or scale up to a fully scaffolded web application.
- **Web Application Hosting** - Enables secure deployment and hosting of Web SDK-based applications within the Cumulocity platform, with built-in support for the [extension ecosystem](#).
- **Microservice SDK** - Offers tools and libraries for developing custom backend services that extend Cumulocity's functionality.
- **Managed Microservice Hosting** - Provides secure, scalable hosting for [custom microservices](#) within the Cumulocity platform infrastructure.
- **Codex** - Comprehensive design system offering UI guidelines, reusable Angular components, and styling utilities to ensure consistent application development.

These capabilities, combined with Cumulocity's built-in [multi-tenancy](#) and [application subscription management](#), provide a robust foundation for scalable application development.

OTHER RESOURCES

- [Open APIs](#) - Well-documented REST APIs that expose complete platform functionality
- [Command Line Interface \(CLI\)](#) - Tools for streamlined development workflows and automation
- Active Developer Community - A collaborative platform for knowledge sharing and problem-solving through the [Tech Community](#)



SYSTEM INTEGRATION

This section guides you to available options and resources for integrating cloud applications and services using various cloud integration platforms. This enables you to automate tasks by connecting cloud applications and services (such as Marketo, Salesforce, Evernote, and Gmail) without writing any code.

The choice of the system integration option depends on the organization's specific needs, technical expertise, and integration requirements. Solutions like Zapier offer flexibility and scalability simplifying integration for non-developers, while self-hosted solutions like n8n.io provide greater control and customization.

By carefully evaluating these options, organizations can select the integration solution that best aligns with their unique requirements, empowering them to unlock the full potential of Cumulocity and drive business transformation.

MICROSERVICES APPROACH - DEVELOPER-FRIENDLY INTEGRATION

For developers seeking a versatile and flexible integration solution, Cumulocity's microservices approach offers a developer-friendly solution. This method involves creating microservices that connect Cumulocity to external systems, allowing for customized data exchange and automation. While requiring initial development effort, the microservices approach provides greater control and adaptability. The option is utilized by many customers for integrating with CRM, ERPs, and FSM tools.

This [article](#) explains how to set up a Cumulocity microservice project in Java, and a corresponding webinar can be found on [YouTube](#).

WORKFLOW AUTOMATION TOOLS

These tools provide a user-friendly interface and extensive integrations, suitable for basic to moderately complex workflows. Popular options include Zapier, n8n.io, and Make.com (formerly Integromat). While some tools cater to straightforward automation tasks, others offer high customizability, open-source access, and self-hosting capabilities, making them suitable for more technically proficient users.

Here are some examples of what you can achieve with these tools:

Zapier

This [article](#) explains how to integrate Cumulocity with Google Sheets to automatically export sensor data for further analysis.

n8n.io

This [article](#) explains how to create a multi-step automation for Cumulocity alarms, generating Zendesk tickets and SMS notifications based on severity.

